

North Campus Open Space bird community analysis: survey years 1-8 (2017-2025)

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Abstract (Executive summary)

Introduction

Methods

Data collection

[Fill this in based on previous documentation in NCOS annual reports and in Dryad data publication]

Data cleaning

- summarize what happens in cleaning script
- standardize species names according to XYZ (specific ebird taxonomy version)

Data analysis

Species richness

Species composition (NMDS & guild trends)

Temporal trends for specific groups

-waterfowl, breeding birds (April-July, evidence of breeding in SB BBS), raptors...

Basic comparison to ebird data over same period

Results

Data set overview

Species richness over time

Comparison to ebird data

[1] "Short-billed Gull"	"Vermilion Flycatcher"
[3] "White-throated Sparrow"	"Horned Grebe"
[5] "American Herring Gull"	"Snowy Plover"
[7] "Band-tailed Pigeon"	"Brewer's Sparrow"
[9] "Lesser Nighthawk"	"Palm Warbler"
[11] "American Barn Owl"	"Golden Eagle"
[13] "American White Pelican"	"Common Gallinule"
[15] "Common Goldeneye"	"Lawrence's Goldfinch"
[17] "Lesser Scaup"	"Brown Pelican"
[19] "Brandt's Cormorant"	"Grasshopper Sparrow"
[21] "Royal Tern"	"Yellow-crowned Night Heron"
[23] "Black-and-white Warbler"	"Black-throated Gray Warbler"
[25] "Western Cattle-Egret"	"MacGillivray's Warbler"
[27] "Nashville Warbler"	"Prairie Falcon"
[29] "Red Crossbill"	"Eurasian Wigeon"
[31] "Black-legged Kittiwake"	"Orchard Oriole"
[33] "Red-throated Loon"	"Western Warbling Vireo"
[35] "Least Tern"	"Semipalmated Sandpiper"
[37] "Baird's Sandpiper"	"Rufous-crowned Sparrow"
[39] "Swainson's Hawk"	"White-winged Dove"
[41] "Brant"	"Lucy's Warbler"
[43] "Hermit Warbler"	"Pine Siskin"
[45] "Olive-sided Flycatcher"	"Pelagic Cormorant"
[47] "Common Merganser"	"Black Turnstone"
[49] "Wood Duck"	"American Redstart"
[51] "Sage Thrasher"	"Bobolink"
[53] "Swamp Sparrow"	"Tennessee Warbler"
[55] "Surf Scoter"	"Vesper Sparrow"
[57] "Reddish Egret"	"Stilt Sandpiper"
[59] "Eastern Kingbird"	"Lapland Longspur"
[61] "Swinhoe's White-eye"	"Iceland Gull"
[63] "Elegant Tern"	"Pacific Loon"
[65] "Red Phalarope"	"Bald Eagle"
[67] "Common Loon"	"Yellow-breasted Chat"

[69] "Tufted Duck"

"Neotropic Cormorant"

[1] "Common Gull"

Species composition over time

The lowest richness occurs at survey year 1 with the following years appearing more similar.

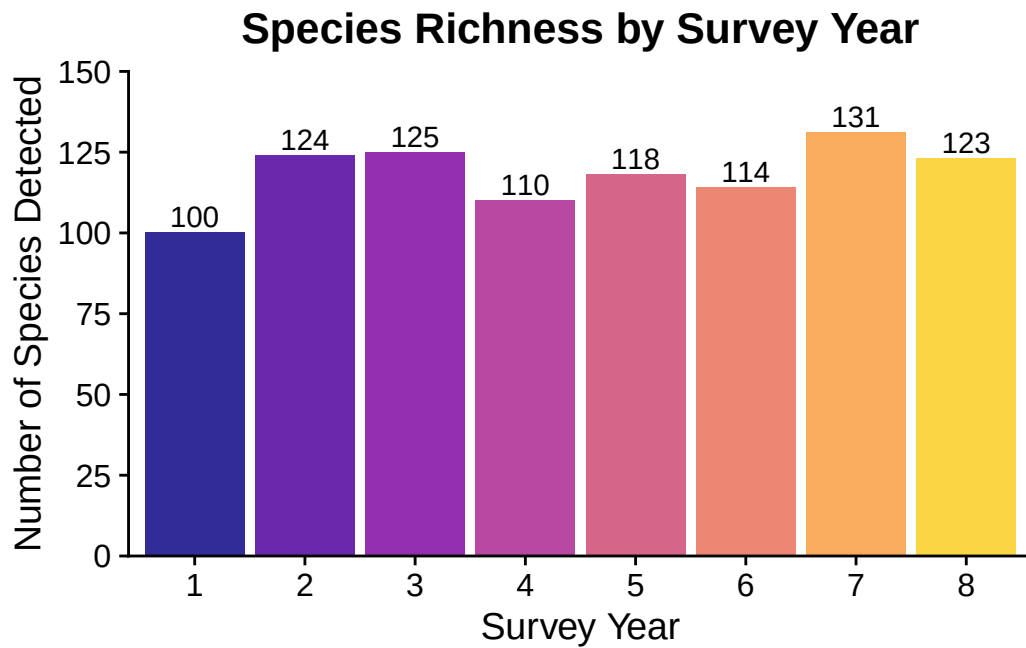


Figure 1: Species richness per survey year at NCOS. The lowest richness occurs at survey year 1 with the rest appearing more similar.

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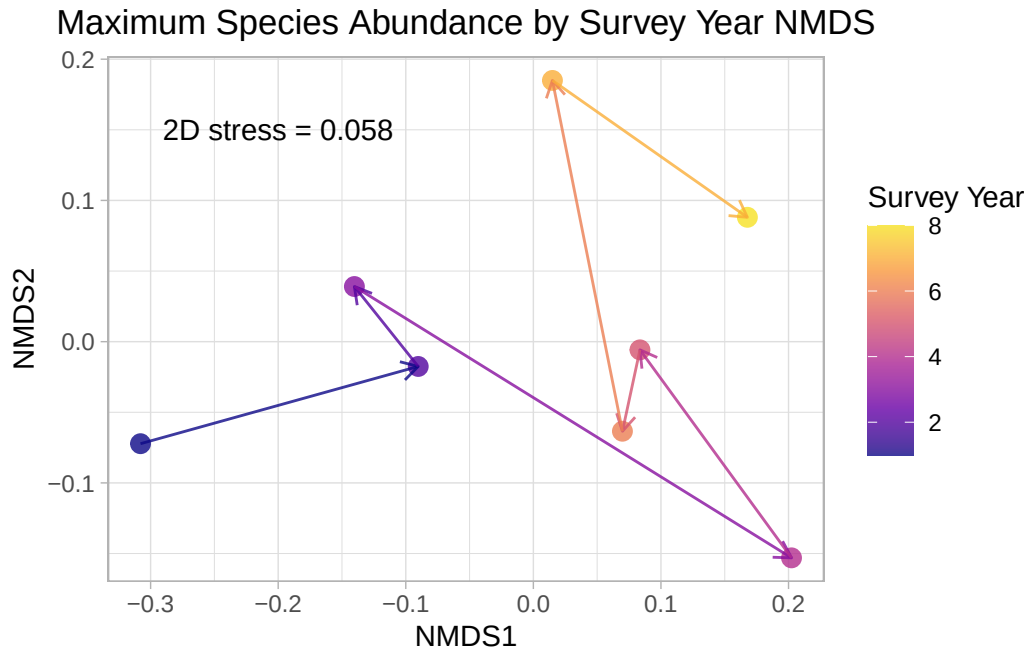
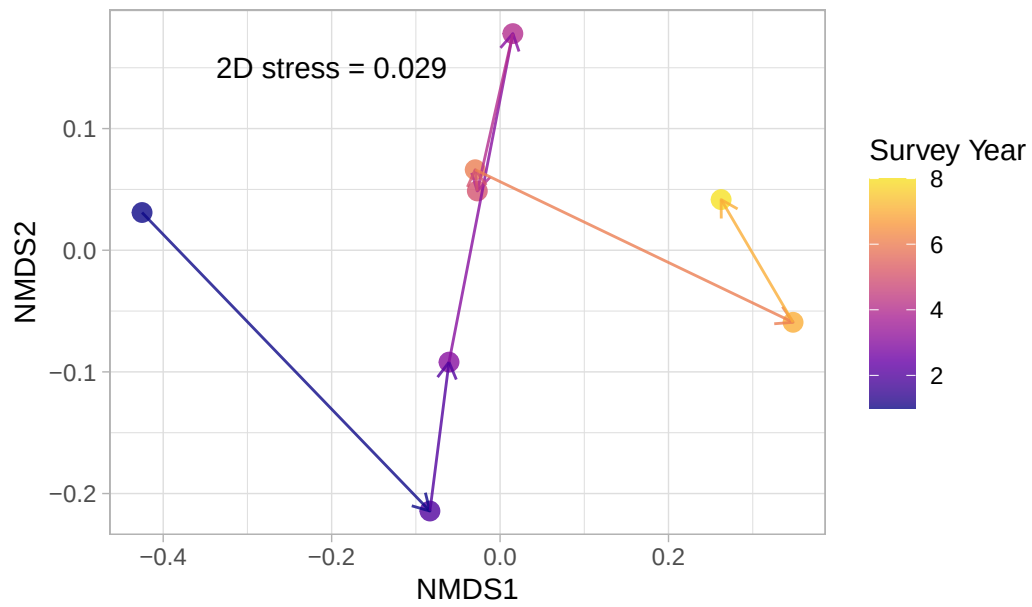


Figure 2: Revised NMDS based on ‘Maximum Species Abundance’ of each species in each year. I reproduced it thinking that with cleaner data it might appear differently, giving insight into species identity similarities or differences, following the bar chart of species richness. The usage of `setdiff()` may be more desirable in this situation.

NMDS based on Jaccard distance (presence/absence)



Discussion